



DoCircuits, an online schematic editor

is aptly named SOLVE, an acronym for students online lab for virtual experimentation. NITK, Surathkal, alone has 70,000 users from all over the world at its virtual website (solve.nitk.ac.in), including a few thousand from Pakistan.

While simulation based labs are set up based on a mathematical base, there are also experiment based labs where experiments are conducted and students are given a feel of the data.

Third type is the remote lab, where an actual lab is made available to users through any device connected through the Internet, iPhone, Android or PC.

"Essentially, we are providing a take-home lab where we share costly laboratory equipment through the remote lab infrastructure," explains Dr K.V. Gangadharan, professor and head of mechanical engineering, NITK, Surathkal.

We have around 100 labs in simulation on the virtual lab website, of which 30 are remote-triggered labs. A remote-triggered lab is an actual experiment at one location to which students can make changes remotely as they interact with the experiment.

The virtual lab provides a complete learning management system, where working principles are taught

and knowledge is imparted through textual and video education, which students can apply to their experiments. Various areas covered by these labs include FPGAs and basic electronics. For instance, IIT Delhi has a virtual wireless lab that gives students actual experiments to learn from and keep modifying antennae available to check propagation testing.

Colleges that are interested in accessing this impressive tool can contact the closest college and sign an expression of interest. There are no monetary requirements, and the only facilities needed are to have 25 computers with the Internet facility and decent bandwidth. The existing team will train the faculty and conduct demonstrations. The Centre for System Design is a centre of excellence (CoE) at NITK, Surathkal, which hosts SOLVE Virtual Lab (solve.nitk.ac.in). The national portal for virtual lab is www.vlab.co.in, where about 100 virtual labs are ready to use.

Virtual electronics design through apps

Autodesk has also brought out their own free tool in the form of Autodesk 123D Circuits. Allowing users to design, compile and simulate electronics projects for free without burning

your Arduino, it is a great way to get started with electronics design.

An interesting feature of this tool is that it allows users to choose from breadboard, schematic or PCB editor views while working on the design.

In addition to that, the editor allows multiple users to work together as a team, thus making it more fun. You can access this tool at 123dapp.com, where you can also find similar tools for creating 3D models and mashups.

Indian tools are equally amazing in this area. For instance, DoCircuits is an amazing tool built by Bengaluru

based Sparsha Learning Technologies Pvt Ltd. It allows you to build circuits, and then test and measure these through lab equipment in a safe simulated environment. Their website docircuits.com allows you to build your first circuit immediately without having to register or go through any of those hassles related with online tools.

If you search deep enough, you can also find browser based design tools that have component supplier integration built into these. For instance, PartSim.com allows you to use its SPICE simulator, AC/DC/transient simulations and waveform viewers to design schematics, which would later be converted into a bill of materials (BOM) by integration with electronic components distributor, Digi-Key.

One particular tool that has received a lot of media attention is CircuitLab, which has been complimented for its user interface and other features like smarter connection lines to connect components, higher-definition schematics and symbol creation, apart from its simulation, plotting and support for behavioural sources and expressions through programmable algebraic sources, which means that you get to play around with programmable voltage and current sources.